LIVER PATIENT
DATA ANALYSIS

A Comprehensive Clinical Analytics Report

SQL & Python Analysis | 583 Patients | 11 Biomarkers

Prepared by Kumar Gaurav

Tools: SQL Server | Python (Pandas, Matplotlib)

Dataset Overview

Indian Liver Patient Dataset — 583 records with 11 clinical biomarkers

Total Patients
583
Records in dataset

Liver Disease
416
71.4% of patients

No Liver Disease
167
28.6% of patients

Average Age
44.7
Years — peak risk cohort

Male Patients
441
75.6% male-dominated

Female Patients
142
24.4% of dataset

Biomarkers
11
Clinical lab parameters

Missing Values
0
Complete dataset

Columns in Dataset

Column	Type	Description
Age / Gender	INT / VARCHAR	Patient demographics
TB / DB	FLOAT	Total & Direct Bilirubin
Alkphos	INT	Alkaline Phosphatase
Sgpt / Sgot	INT	Liver Enzymes (ALT/AST)
TP / ALB	FLOAT	Total Protein & Albumin
AG_Ratio	FLOAT	Albumin-Globulin Ratio
Selector	VARCHAR	Diagnosis Label

Insight

The dataset is complete with no missing values — ideal for direct analysis. The 416:167 imbalance (liver disease dominant) must be considered when evaluating model accuracy. 75.6% male representation reflects known gender bias in liver disease prevalence.

SQL Analysis — Basic Level (Q1–Q5)

Understanding the Data: Record Counts, Demographics & Disease Distribution

SQL *Q1: Total Records*

```sql
SELECT
    COUNT(*) as total_records
from
    liver_patient_dataset;
```

Result: 583 patients total in dataset

Complete records — no incomplete rows

SQL *Q2: Male vs Female*

```sql
select
    gender,
    COUNT(gender) as total_count
from liver_patient_dataset
group by Gender;
```

Male: 441 (75.6%)

Female: 142 (24.4%)

SQL *Q3 & Q4: Age Statistics*

```sql
SELECT
    AVG(age) average_age
from
    liver_patient_dataset;
```

Avg Age

44.7

years

Min Age

4

youngest

Max Age

90

oldest patient

SQL *Q5: Disease Distribution*

```sql
select
    top 5 *
from liver_patient_dataset
select
    Selector,
    COUNT(Selector) as total_count
from liver_patient_dataset
group by Selector;
```

Insight

Key Takeaway: 71.4% of patients have liver disease — the dataset skews toward positive cases. Male patients are 3× more common than female. Average patient age of 44.7 puts most cases in working-age adults (41–60 range). Age range spans 4–90 years, indicating population-wide impact.

SQL Analysis — Intermediate Level (Q6–Q10)

Aggregation & Filtering: Bilirubin, Enzymes, Age Groups & Albumin

SQL *Q6: Avg Bilirubin by Diagnosis*

```sql
select
    selector,
    AVG(TB) as average_TB
from
    liver_patient_dataset
group by
    selector;
```

SQL *Q8: Avg Enzyme Levels by Gender*

```sql
SELECT
    gender,
    AVG(sgpt) as average_sgpt,
    AVG(sgot) as average_sgot
from
    liver_patient_dataset
group by
    gender
```

SQL *Q7: Age Group Distribution*

```sql
select
    case
        when age between 0
        and 20 then '0-20'
        when age between 21
        and 40 then '21-40'
        when age between 41
        and 60 then '41-60'
        else '60+'
    end as age_group,
    selector,
    COUNT(*) total_patients
from
    liver_patient_dataset
group by
    case
        when age between 0
        and 20 then '0-20'
        when age between 21
        and 40 then '21-40'
        when age between 41
        and 60 then '41-60'
        else '60+'
    end,
    Selector
order by
    total_patients desc;
```

SQL *Q9: Abnormal Albumin (ALB < 3.5 or > 5.0)*

Abnormal Albumin

372

63.8% of all patients

SQL *Q10: Top 5 Alkphos*

```sql
SELECT TOP 5 * FROM liver_patient_dataset
ORDER BY Alkphos DESC;
-- Max observed: 2110 (normal <120)
```

Insight

Liver disease patients have 3.6× higher bilirubin (4.16 vs 1.14 mg/dL) — a critical diagnostic marker. The 41–60 age group has the highest case count. Male enzyme levels are ~65% higher than female, suggesting gender-specific metabolic differences. 63.8% abnormal albumin signals widespread liver dysfunction.

SQL Analysis — Advanced Level (Q11–Q15)

Insights & Pattern Recognition: Alcohol Risk, Protein Profiles & Predictive Rules

SQL *Q11: Alcohol-Related Liver Risk (AST/ALT Ratio)*

```
SELECT
    Selector,
    CASE
        WHEN (Sgot / NULLIF(Sgpt, 0)) > 2 THEN 'High (Alcohol-related)'
        WHEN (Sgot / NULLIF(Sgpt, 0)) BETWEEN 1
        AND 2 THEN 'Moderate'
        ELSE 'Low'
    END AS risk_category,
    COUNT(*) AS total_patients
FROM
    liver_patient_dataset
GROUP BY
    Selector,
    CASE
        WHEN (Sgot / NULLIF(Sgpt, 0)) > 2 THEN 'High (Alcohol-related)'
        WHEN (Sgot / NULLIF(Sgpt, 0)) BETWEEN 1
        AND 2 THEN 'Moderate'
        ELSE 'Low'
    END;
```

Metric	Liver Disease	No Liver Disease	Clinical Normal
Total Protein (TP)	6.46 g/dL	6.54 g/dL	6.3–8.2
Albumin (ALB)	3.06 g/dL	3.34 g/dL	3.5–5.0
A/G Ratio	0.91	1.03	1.1–2.5

Flagged Patients

6

TB>5 yet no liver disease

These patients warrant re-evaluation — elevated bilirubin without diagnosis could indicate early-stage disease or comorbidity.

SQL *Q15: Rule-Based Prediction Accuracy*

Rule: IF (TB>1.2 + SGPT>40 + ALB<3.5) >= 2 → Liver Disease
Total: 583 | Correct: 371 | Accuracy: 63.64%
True Positives: 238/416 | True Negatives: 133/167

Insight

Protein markers are consistently lower in liver disease patients: ALB drops 8.4% and A/G ratio drops 11.7% vs healthy — strong indicators. Moderate AST/ALT ratio (1–2) is most prevalent across both groups. 6 high-bilirubin false-negative patients need urgent clinical review. The 3-biomarker rule achieves 63.64% accuracy — promising for a simple heuristic.

SQL Analysis — Advanced Level (Q11–Q15)

SQL *Q12: Protein Levels — Disease vs No Disease*

```
select
    selector,
    ROUND(AVG(tp), 2) as total_protien,
    ROUND(AVG(alb), 2) as total_albumin,
    ROUND(AVG(ag_ratio), 2) as total_ag_ratio
from
    liver_patient_dataset
group by
    Selector;
```

SQL *Q13: High Bilirubin but No Liver Disease (False Negatives)*

```
select
    *
from
    liver_patient_dataset
where
    tb > 5.0
    AND selector = 'No Liver Disease';
```

SQL Analysis — Advanced Level (Q11–Q15)

SQL *Q14: Most Common Feature Combination*

```
SELECT
    TOP 1 Gender,
    CASE
        WHEN Age BETWEEN 0
        AND 20 THEN '0-20'
        WHEN Age BETWEEN 21
        AND 40 THEN '21-40'
        WHEN Age BETWEEN 41
        AND 60 THEN '41-60'
        ELSE '60+'
    END AS age_group,
    CASE
        WHEN Sgpt > 40 THEN 'High SGPT'
        ELSE 'Normal SGPT'
    END AS sgpt_level,
    COUNT(*) AS total_patients
FROM
    liver_patient_dataset
WHERE
    Selector = 'Liver Disease'
GROUP BY
    Gender,
    CASE
        WHEN Age BETWEEN 0
        AND 20 THEN '0-20'
        WHEN Age BETWEEN 21
        AND 40 THEN '21-40'
        WHEN Age BETWEEN 41
        AND 60 THEN '41-60'
        ELSE '60+'
    END,
    CASE
        WHEN Sgpt > 40 THEN 'High SGPT'
        ELSE 'Normal SGPT'
    END
ORDER BY
    total_patients DESC;
```

Python Analysis — EDA & Data Profiling

Pandas-based exploration: Shape, dtypes, missing values, distributions

```
# Count how many patients have liver disease vs no liver disease.
df['Selector'].value_counts()
```

```
# Check the shape(rows,colums) of the dataset.

df.shape
```

```
# Check for missing values in each column.
df.isnull().sum()
```


```
# Count how many patients have liver disease vs no liver disease.
df['Selector'].value_counts()
```


```
# Find the average age of patients.
df['Age'].mean()
```

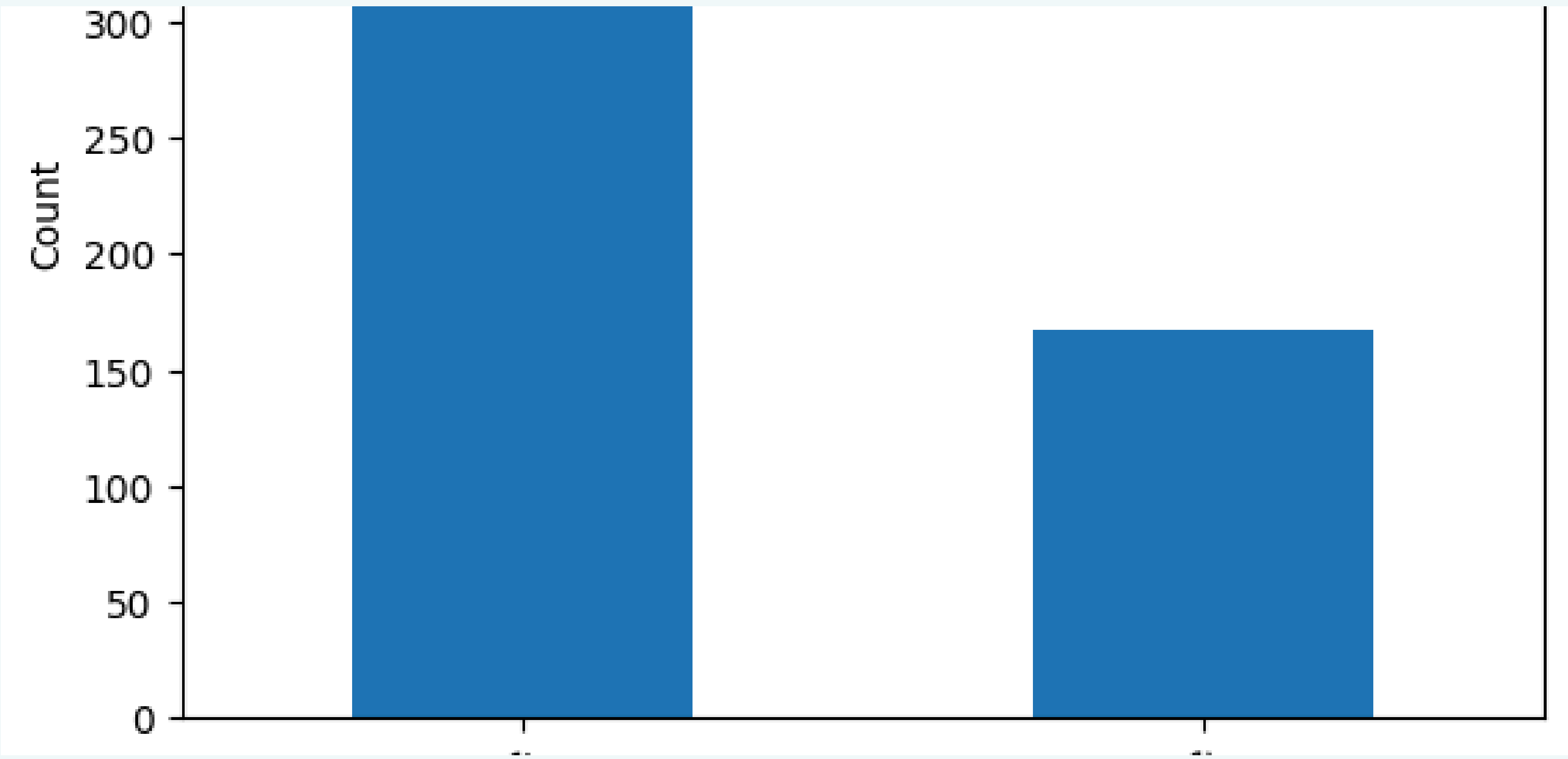
```
... np.float64(44.74614065180103)
```

```
# Calculate the average total bilirubin(TB) for liver disease vs non-disease patients.

result = df.groupby('Selector')['TB'].mean()
print(result)
```

```
... Selector
Liver Disease       4.164423
No Liver Disease    1.142515
Name: TB, dtype: float64
```

Disease Distribution



Insight

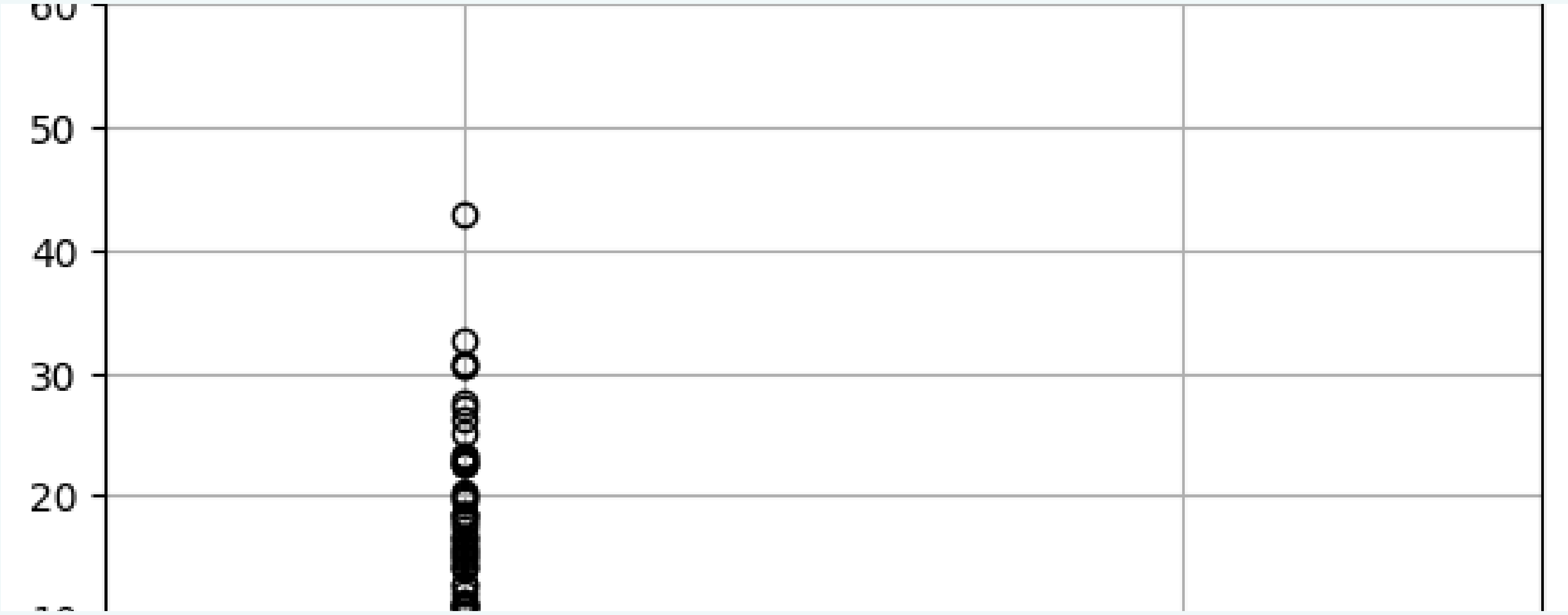
Python confirms: 583 records, 11 columns, 0 missing values — a clean, production-ready dataset. The 416:167 class imbalance (liver disease vs healthy) will impact ML model performance and requires attention. Pandas profiling takes seconds and validates what SQL already discovered.

Python Analysis — Bilirubin, Age Groups & Albumin

Groupby analysis, age binning, abnormal albumin detection

Total Bilirubin Distribution

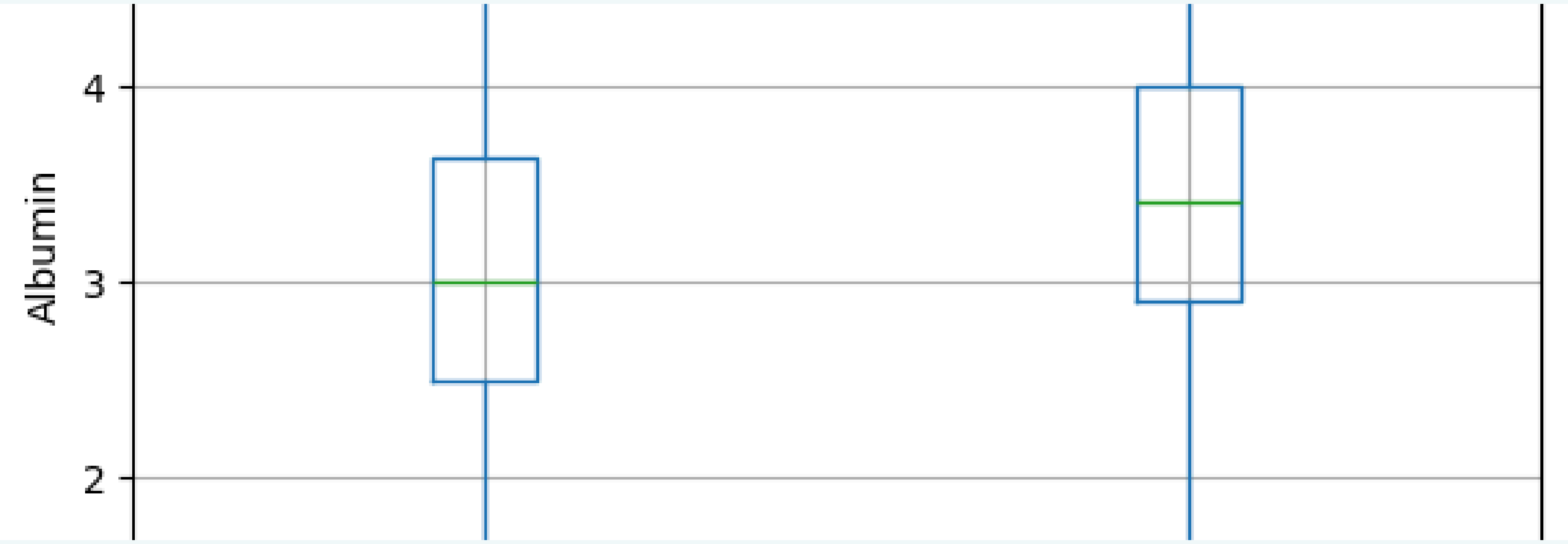
TB by Disease Status



Code: TB by Selector

Code: Age Groups

Albumin by Disease Status



Code: Abnormal Albumin

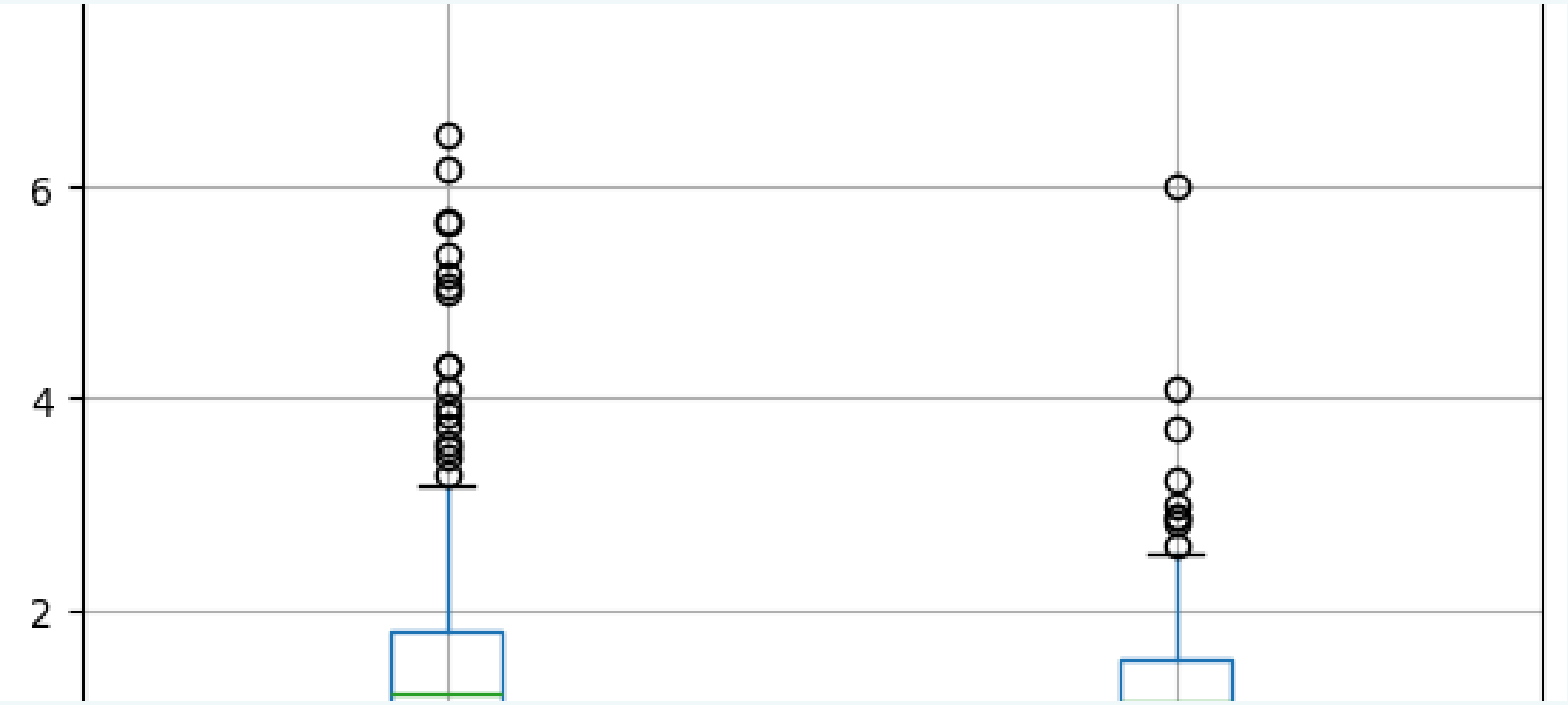
Insight

Bilirubin is right-skewed — most patients cluster below 5 mg/dL, but disease patients show dramatically elevated TB values (up to 75 mg/dL). Albumin box plots clearly separate disease vs healthy patients: liver disease patients have lower median albumin (3.06 vs 3.34 g/dL) with more outliers. The 41–60 age bracket carries the highest disease burden (242 patients).

Python Analysis — Advanced: AST/ALT Ratio & Prediction

Alcohol-related risk profiling, protein comparison & rule-based ML

AST/ALT Ratio by Disease Status



Code: AST/ALT Ratio

Code: Protein Comparison

Code: Rule-Based Prediction

Confusion Matrix

Predicted →	Liver Disease	No Disease
Actual: Liver Disease	**238 ✓**	178 ✗
Actual: No Disease	34 ✗	**133 ✓**

Insight

Albumin and A/G ratio are the most discriminating protein markers between disease groups. Disease patients show consistently lower values across all three protein metrics (TP, ALB, AG_Ratio). Even a 3-variable rule captures meaningful signal — further improvement possible with ML algorithms like Random Forest or XGBoost.

Insight

The AST/ALT ratio is higher in liver disease patients (1.49 vs 1.32), consistent with alcohol-related and hepatocellular damage patterns. The 3-biomarker rule (TB + SGPT + ALB) achieves 63.64% accuracy — good for a simple heuristic but needs improvement. The model correctly identifies 57.2% of actual disease cases (sensitivity), suggesting strong recall opportunity with additional features.

Key Clinical Insights — Combined SQL & Python Findings

What the data tells us about liver disease patterns

01 Bilirubin is the Strongest Marker

Liver disease patients show 3.6× higher average total bilirubin (4.16 vs 1.14 mg/dL). High TB alone correctly classifies a large proportion of cases.

02 Albumin Signals Liver Dysfunction

63.8% of all patients have below-normal albumin (<3.5 g/dL). Disease patients average ALB of 3.06 vs 3.34 in healthy patients — a consistent, measurable gap.

03 41–60 Age Group is the High-Risk Cohort

The 41–60 bracket has the most liver disease cases (185 patients). Combined with male gender dominance (75.6%), older males are the highest-risk demographic.

04 Gender Gap in Enzyme Levels

Male patients have ~65% higher SGPT/SGOT enzyme levels than female patients. This may reflect higher alcohol consumption, occupational exposure, or biological differences.

05 AST/ALT Ratio Identifies Alcohol Risk

Liver disease patients show higher AST/ALT ratios (avg 1.49). A ratio >2 is a recognized clinical indicator of alcohol-related liver disease.

06 6 High-Risk Undiagnosed Patients

6 patients have TB >5 mg/dL yet are labeled 'No Liver Disease' — possible misdiagnosis or early-stage disease. These cases require urgent clinical re-review.

Insight

Overall Assessment: The dataset provides strong, actionable clinical signals. Bilirubin, albumin, and the AST/ALT ratio together form a reliable diagnostic triad. The 3-biomarker rule achieves 63.64% accuracy — indicating significant room for improvement with machine learning. Gender and age remain critical demographic risk factors that should be embedded in any predictive model.

Clinical & Analytical Recommendations

What to do — and what to avoid — based on the data analysis

RECOMMENDED ACTIONS

Use TB + ALB + SGPT as Primary Triage Biomarkers

These three markers together achieve 63.64% rule-based accuracy. Prioritize them for early screening protocols.

Implement Age-Gender Risk Stratification

Males aged 41–60 are the highest-risk group. Design targeted screening programs for this demographic.

Review 6 High-Bilirubin 'No Disease' Cases

Patients with TB>5 but no diagnosis may have early-stage disease. Schedule follow-up diagnostics immediately.

Deploy Machine Learning for Improved Accuracy

Upgrade from 63.64% rule-based accuracy using Random Forest, XGBoost, or Logistic Regression with cross-validation.

Address Class Imbalance Before ML Modeling

Use SMOTE, oversampling, or cost-sensitive learning to handle the 416:167 class imbalance before training models.

Track AST/ALT Ratio for Alcohol Risk Monitoring

Routinely calculate AST/ALT ratio (>2 = high risk). Flag high-ratio patients for lifestyle intervention programs.

WHAT TO AVOID

Don't Rely on Single Biomarkers for Diagnosis

No single marker alone is sufficient. Always use a combination of TB, ALB, SGPT, and demographic factors.

Don't Ignore Class Imbalance in Predictions

With 71.4% positive cases, a naive model predicting all patients as 'diseased' achieves 71% accuracy — misleadingly high.

Don't Assume Female Patients are Low-Risk

Lower enzyme levels in females may mask underlying disease. Gender-specific thresholds should be established.

Don't Use the 3-Rule Model for Final Diagnosis

63.64% accuracy means ~36% errors. This rule is for screening only — not clinical decision-making or treatment planning.

Don't Overlook Extreme Alkphos Values

Top patients have Alkphos >2000 (normal: <120). These extreme outliers may indicate obstructive biliary disease — investigate separately.

Don't Ignore the 6 High-Bilirubin Anomalies

False negatives with TB>5 are a clinical liability. Automated alerts should flag these cases before discharge or sign-off.

Priority Action: Implement a 3-step clinical workflow — (1) Screen using TB + ALB + SGPT triage rule, (2) Flag anomalies for radiological follow-up, (3) Deploy ML model with minimum 80% sensitivity threshold before clinical rollout.

THANK YOU

Liver Patient Analytics Report

- 583 Patients Analyzed
- 15 SQL Queries Executed
- 14 Python Operations Performed
- 5 Visualization Charts Generated
- 63.64% Prediction Rule Accuracy

Prepared by Kumar Gaurav

Tools Used: SQL Server | Python | Pandas | Matplotlib

Analysis Framework: Exploratory → Diagnostic → Predictive